

Multi-Agent Intelligence System (Google ADK)

Project Objectives (Should be completed in 2 weeks)

- 1. Design and implement a multi-agent system using Google ADK.
- 2. Integrate RAG (Retrieval-Augmented Generation) for knowledge management.
- 3. Implement web search capabilities for real-time information.
- 4. Build custom tools demonstrating agent composition.
- 5. Add observability/monitoring for production insights.
- 6. Deploy to Google Cloud Platform (Cloud Run).
- 7. Present solution via code, presentation (ppt), and demo video.

Core Requirements

1. Google ADK Multi-Agent System (40%)

System must include minimum 3 specialized agents: Orchestrator, Researcher, and Reviewer/QA.

Communication Patterns (Implement 2+): Sequential Flow, Parallel Execution, Hierarchical Delegation, Feedback Loop.

State Management: Session management, state sharing, and memory.

2. RAG Support (25%)

Pipeline: PDF extraction, semantic chunking, Vertex AI text-embedding-005, and FAISS vector storage.

3. Web Search Capabilities (15%)

API Choice: Tavily API. Agents must intelligently choose between RAG and Web Search based on the query type.

Project Use Case Examples

System Type	Primary Knowledge Base	Web Search Use
Research Assistant	Academic papers, journals	Latest citations
Customer Support	Product manuals, FAQs	Competitor pricing
Legal Analyst	Contracts, regulations	Recent court decisions
E-Commerce Bot	Product catalogs	Product reviews

Sample Project Structure

Multi-Agent Intelligence System (Google ADK)

```
student_project/
|-- src/
|   |-- agents/          # Base, Research, and Coordinator agents
|   |-- rag/             # Document processing and Vector DB logic
|   |-- tools/           # Web search and custom analysis tools
|   |-- observability/   # Logging, metrics, and tracing
|   |-- main.py          # Entry point
|-- knowledge_base/      # Source documents (PDFs/Text)
|-- notebooks/           # Exploration and testing scripts
|-- tests/               # Unit and integration tests
|-- deployment/          # Docker and Cloud Run configurations
|-- requirements.txt      # Python dependencies
```